Analog and Digital Electronics (ESC-301)

Course Details

Credit Points: 3

Pre-Requisites

1. Basic Electronics Parts I & II learned in the First year, semesters 1 & 2. Basic BJTs.
2. Basic concept of the working of P-N diodes, Schottky diodes.
3. Basic FETs and OPAMP as a basic circuit component. Concept of Feedback.

Syllabus Outline

Unit 1

- Different Classes of Amplifiers (Class-A, B, AB and C – basic concepts, power, efficiency)
- Recapitulation of basic concepts of Feedback and Oscillation
- Phase Shift, Wein Bridge oscillators
- Astable & Monostable Multivibrators
- Schmitt Trigger circuits, 555 Timer.

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Unit 2

- Binary Number System & Boolean Algebra (recapitulation)
- BCD, ASCII, EBCDIC, Gray codes and their conversions
- Signed binary number representation with 1's and 2's complement methods, Binary arithmetic, Venn diagram, Boolean algebra (recapitulation)
- Representation in SOP and POS forms; Minimization of logic expressions by algebraic method.
- Combinational circuits: Adder and Subtractor circuits (half & full adder & subtractor)
- Encoder, Decoder, Comparator, Multiplexer, De-Multiplexer and Parity Generator.

Unit 3

- Sequential Circuits: Basic Flip-flop & Latch
- Flip-flops: SR, JK, D, T and JK Master-slave Flip Flops
- Registers: SISO, SIPO, PIPO, PISO
- Ring counter, Johnson counter
- Basic concept of Synchronous and Asynchronous counters (detailed design of circuits excluded)
- Design of MOD N Counter.

Unit 4

- A/D and D/A conversion techniques – Basic concepts (D/A: R-2-R only, A/D: successive approximation)
- Logic families: TTL, ECL, MOS, and CMOS – basic concepts.

Analog and Digital Electronics Lab (ESC-391)

LABORATORY EXPERIMENTS

Analog Electronics

- Design a Class A amplifier
- Design a Phase-Shift Oscillator
- Design of a Schmitt Trigger using 555 timers

Digital Electronics

- Design a Full Adder using basic gates and verify its output / Design a Full Subtractor circuit using basic gates and verify its output.
- Construction of simple Decoder & Multiplexer circuits using logic gates.
- Realization of RS / JK / D flip flops using logic gates
- Design of Shift Register using J-K / D Flip Flop
- Realization of Synchronous Up/Down counter
- Design of MOD- N Counter
- Study of DAC

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